What is Multicollinearity?

Multicollinearity occurs when two or more independent variables in a dataset are **highly correlated**, meaning that one variable can be **linearly predicted** from the others. This leads to redundancy in the model and affects statistical analysis, especially in linear regression.

Why this effect some model performance?

The model struggles to determine the effect of each variable, leading to large and unreliable coefficients**.**

y(independent) = b1x1 + b2x2 + b3

It will not be able to find b1,b2,b3 coefficients efficiently

How to Detect It?

1. **Correlation Matrix** – Check high correlation values (>0.8) between independent feature

Models Severely Affected (Major Problem)

1. **Linear Regression (OLS, Ridge, Lasso, ElasticNet)** – Leads to **unstable coefficients** and poor interpretability.
2. **Logistic Regression** – Affects **coefficient estimates**, making the model unreliable.
3. **Support Vector Machine (SVM) with Linear Kernel** - High correlation affects hyperplane orientation
4. **Naïve Bayes** – Assumes independence of features, so correlated predictors can distort probability estimates

Models That Are Not Affected (No Problem)

8. **Decision Trees (CART, ID3, C4.5, etc.)** – Splits data based on feature importance, so multicollinearity has minimal impact.
9. **Random Forest** – Uses multiple trees, so it naturally reduces feature redundancy.
10. **Gradient Boosting (XGBoost, LightGBM, CatBoost)** – Similar to random forests, not affected by multicollinearity.

11. **Neural Networks (Deep Learning)** – Can learn non-linear feature interactions and ignore redundant features.
12. **K-Nearest Neighbors (KNN)** – Distance-based model; correlated features won't significantly impact predictions.

HOW TO SOLVE MULTICOLLINEARITY PROBLEM?

1.Remove one of the correlated variables

2.Use Principal Component Analysis (PCA) to reduce dimensionality